Substituting into gives

From ,

The remaining argument is the same as in “Method ”.
Scoring: see “Method ”.

V. (points)
Ambient pressure is . Initially the air has pressure , temperature , volume .
Contact with the iceberg brings equilibrium at temperature and volume . This is an isobaric process, so

The change in the air's internal energy during this process is

The work done by the atmosphere on the air is

Let be the heat from the iceberg to the air. The first law of thermodynamics gives

The four preceding equations give

In , is negative: heat actually flows from the air to the iceberg.
After cooling to the ice temperature, the air is heated at fixed volume , from to ambient temperature , absorbing heat from its surroundings. If pressure is ,

Let be the nozzle volume. The first jet lowers vessel pressure from to . The given conditions imply